Optics Letters

Optical anapoles excited by UV-A illumination

Diogo Poeta,[1,2,†] Rafael Vilarinho,[1,†] Martín López-García,[3] and Pieter A. A. De Beule[1,*]

[1] International Iberian Nanotechnology Laboratory, Av. Mestre José Veiga s/n, Braga 4715-330, Portugal
[2] Biomolecular Photonics Research Group, Faculty of Physics, Bielefeld University, Bielefeld, Germany
[3] Instituto de Óptica, IO-CSIC, Serrano 121, Madrid 28006, Spain
[†] These authors contributed equally to this work.
*pieter.de-beule@inl.int

Received 17 July 2024; revised 4 November 2024; accepted 7 November 2024; posted 11 November 2024; published 11 December 2024

We report the excitation of optical anapole states at ultraviolet (UV) wavelengths. Numerical simulations indicate that TiO_2 nano-rectangles with varying length-to-width ratios can support such modes within the 350–380 nm range. We further propose a two-dimensional periodic arrangement of these nano-rectangles deposited atop a fused silica substrate. Understanding and manipulating optical anapole states in the ultraviolet spectrum is crucial for advancing next-generation photonic devices and enhancing nonlinear optical processes, such as generation of highly energetic vacuum ultraviolet light through third-harmonic generation.

© 2024 Optica Publishing Group under the terms of the Optica Open Access Publishing Agreement

https://doi.org/10.1364/OL.536925

Light scattering of nano-structures is fundamental to how we perceive colors and shapes and essentially determines our visual environment [1]. It namely results in radiation that contributes to the visibility of objects. Scattering depends on both the geometrical properties of the nano-structure and the wavelength and polarization properties of the illuminating beam. One intriguing question that has been addressed is the identification of the geometrical properties of a nano-structure that reduce the overall scattered light intensity to a minimum, thereby creating strong confined electric and magnetic fields inside the nano-structure [2], a quest culminating in the creation of a three-dimensional invisibility cloak [3].

Understanding non-radiating states advanced significantly in 1957 with Yakov Zel'dovich's discovery of parity violation in weak interactions, proving that certain particles exhibit a toroidal dipole moment, eventually referred to as the anapole moment [4]. The concept of an anapole mode was translated to the nanophotonics field, characterized by their ability to confine electromagnetic fields inside and near the nano-structure [5], largely eliminating far-field radiation. This confinement makes the object quasi-invisible at the resonant wavelength of the incident light. This minimizes scattering and absorption, thereby reducing visibility in the far-field.

The family of toroidal moments are retrieved as next-order terms from the expansion of the Cartesian multipoles [6]. For instance [7], we have that the total electric dipole moment $\mathbf{D}$ is given by Eq. (1):

$$\mathbf{D} = \mathbf{p} + \frac{i\left(\frac{2\pi}{\lambda_0}\right)}{c}\mathbf{T}, \tag{1}$$

where $\mathbf{p}$ is the electric dipole moment, $\mathbf{T}$ is the toroidal dipole moment, λ_0 is the wavelength in a vacuum, and c is the speed of light. The condition $\mathbf{D} = 0$, with $|\mathbf{p}| \neq 0$ and $|\mathbf{T}| \neq 0$, is necessary for exciting the electric anapole state [8]. In a real scenario, the total electric dipole moment approaches a minimum value or remains close to zero, indicating proximity to the anapole state, highlighting the significance of electromagnetic phenomena. Exciting anapole states in the ultraviolet (UV) spectrum has potential applications in various fields such as photonics, sensing technologies, and nonlinear optics [9–12]. As a result, it has been increasingly studied for its role in enhancing light–matter interactions and its potential to minimize energy losses in these applications. This distinction highlights that anapole states minimize electromagnetic field scattering, rather than eliminate it.

Figure 1 illustrates various geometries that have been shown to excite anapoles for wavelengths between 550 nm and 2265 nm [9,13–16]. These geometries utilize nano-structures specifically designed to support the formation of toroidal dipole moments, achieving non-radiating resonances by inducing particular electromagnetic field distributions. By leveraging nano-structures with varying aspect ratios, we aim to excite anapole states using ultraviolet (UV) light.

We identified that nano-structures with equal length-to-width are ineffective at exciting anapole states at short wavelengths, particularly within the UV spectrum below 400 nm. This limitation arises from their inability to achieve the necessary field configurations for destructive interference. In contrast, structures with varied aspect ratios and specific geometric modifications were able to support anapole states effectively. This difficulty stems from the limited availability of materials that possess the necessary optical properties at these wavelengths, which are strong refractive index and very low extinction coefficient [17,18]. This constraint complicates the design and realization of anapole states in the UV region, thus demanding appropriate material selection to overcome the inherent limitations posed by the available material properties [12,19].

Our investigation into nano-structures capable of supporting non-radiating anapole states in the UV regime involves a

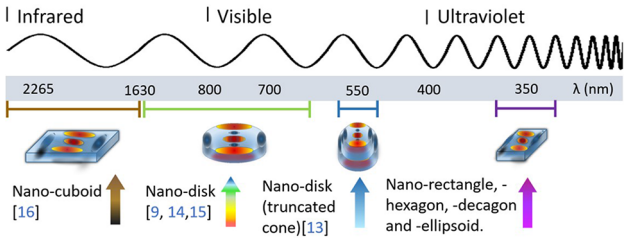


Fig. 1. Illustration of some geometries that are proven of exciting anapoles for wavelengths between 550 nm and 2265 nm. Our goal is to excite anapole states in the UV regime.

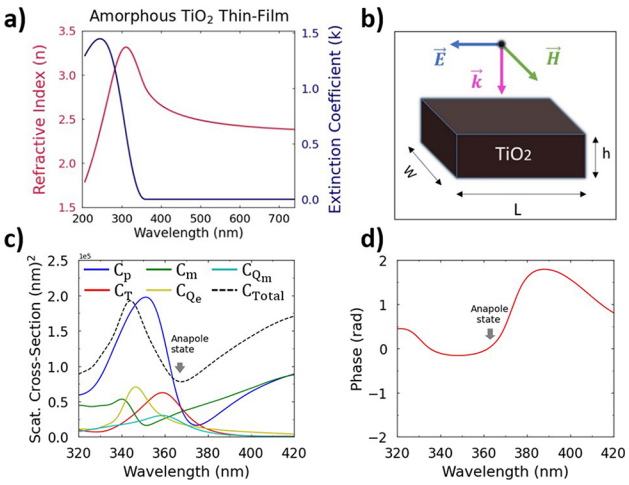


Fig. 2. (a) Dispersion profile of the amorphous TiO_2 refractive index (n) and extinction coefficient (k). (b) Schematic of a free-standing TiO_2 nano-rectangle with the following dimensions: length L = 190 nm, width w = 160 nm, and height h = 100 nm, showing light propagation direction ($\vec{k}$) and polarization direction ($\vec{E}$). (c) Contributions of various multipole moments: C_p (blue), C_T (red), C_m (green), C_{Qe} (yellow), C_{Qm} (cyan blue), and total contribution C_{total} (black). (d) Phase difference between the electric and toroidal dipoles (allusive to Eq. (1)).

two-step approach [20]. First, we select a material with appropriate optical properties for UV operation. Second, we design nano-structures using this material to achieve anapole states at specific UV wavelengths. Amorphous TiO_2 was chosen for this study due to its excellent dielectric properties—specifically, its refractive index (n) and extinction coefficient (k) [21]—which are well-suited for operations at UV wavelengths below 400 nm [22]. As shown in Fig. 2(a), the dielectric properties of this material, according to Jolivet *et al.* (2023) [23], indicate that the material possesses a high refractive index with a low extinction coefficient over a specific range of wavelengths, suggesting that these optical properties are conducive to the excitation of anapoles for wavelengths higher than 350 nm, where the extinction coefficient (k) approaches zero. The selection of TiO_2 was influenced by the findings in Semmlinger *et al.* (2019) [13], where the authors demonstrate the effective utilization of TiO_2 metasurfaces for generating vacuum UV light. Their work highlights the harnessing of anapole resonances within TiO_2 nano-structures, optimizing conversion efficiency at short wavelengths. These characteristics establish TiO_2 as a compelling material for advancing applications in UV light generation [13].

Once we have defined the material of choice as a candidate to operate in the UV, it is necessary to determine which nano-structure geometry is more suitable for the anapole excitation. The selected rectangular shape (Fig. 2(b)) is free-standing rather than positioned on a substrate. In order to probe the anapole excitation, we consider a plane wave incident perpendicular to the rectangular shape, as indicated in Fig. 2(b). The incident electric field ($\vec{E}$) is aligned parallel to the length of the nano-rectangle, which is critical for the excitation of the anapole state due to the polarization-dependent response of the structure. The propagation direction of the incident light is represented by the wave vector ($\vec{k}$), and the induced magnetic field ($\vec{H}$) is depicted orthogonal to $\vec{E}$.

We investigated the scattering characteristics of different nano-structured metasurfaces using the multipole expansion nanophotonics package (MENP) tool developed by Hinamoto and Fujii [24]. This tool, adapted for our study, enabled us to decompose the scattering cross section into partial contributions of multipoles in Cartesian coordinates up to the second order. Specifically, these multipoles include the electric dipole C_p, toroidal dipole C_T, magnetic dipole C_m, electric quadrupole C_{Qe}, magnetic quadrupole C_{Qm}, and total scattering cross section C_{total}. This decomposition offers valuable insight into the complex light–matter interactions within the metasurface. The results from the multipolar decomposition are visually represented in Figs. 2(c) and 2(d), illustrating the electric anapole condition achieved within the nano-rectangle (Section 7 of Supplement 1 provides a brief discussion of the resonance profile). The plot demonstrates the interaction between multipole moments, with the electric anapole condition marked by a significant dip in scattering—a direct consequence of destructive interference between C_p and C_T [25]. This condition is characterized by the cancellation of the electric dipole **p** and toroidal dipole **T** moments. Notably, at approximately 366.5 nm, a pronounced minimum in the total scattering cross section (C_{total}) indicates the occurrence of the electric anapole condition.

Subsequent simulations, using the finite-difference time-domain method (Lumerical FDTD), provided spatially resolved electric and magnetic field distributions (Figs. 3(a) and 3(b), respectively), allowing us to examine the anapole resonance in detail. The electric field is confined within the structure, while the magnetic field distribution shows localized regions of intensity, a characteristic of anapole states. This characteristic ensures that the resonance is an anapole, a non-radiating source, as it minimizes electromagnetic fields outside its excitation region [26]. Additionally, the spectral position of the anapole resonance within the specified UV wavelength range is demonstrated to shift by varying the width of the free-standing nano-rectangle. The results highlight the adaptability of the design, showing a direct correlation between the width of the resonator and its optical response. A similar relationship was observed by Li *et. al* [27]. Notably, there is no data for a width of 190 nm (see Fig. 3(c)), as there is no anapole excitation for this size. This reinforces the need for specific aspect ratios of length and width to excite anapoles in the UV range.

The anapole state's spectral position remains constant for incident angles up to 25° (Fig. S1 in Supplement 1), suggesting robustness of the meta-atom, important for applications where light incidence is variable or non-perpendicular. Beyond this angle, the scattering increases and a minimum in the total scattering cross section is no longer observed. This stability ensures

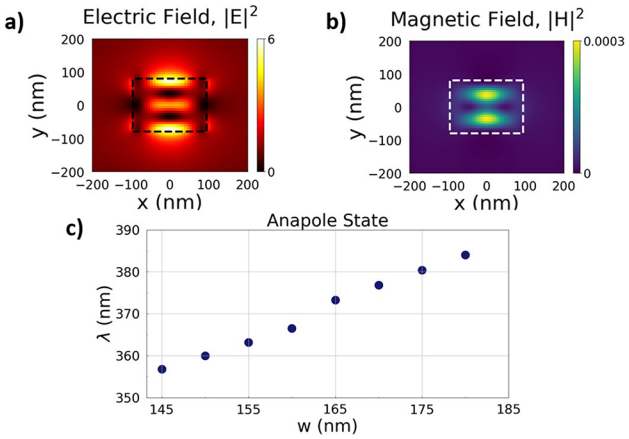


Fig. 3. (a) Electric and (b) magnetic field intensities in the free-standing nano-rectangle structure. (c) Shift of the anapole wavelength when varying the width (w) of the nano-rectangle. The electric and magnetic fields are normalized relative to the source.

that the desired anapole resonance is preserved even in experimental scenarios where perfect alignment cannot be guaranteed. Temporal analysis (Figs S2 and S3 in Supplement 1) reveals the anapole state undergoes distinct stages of formation and decay. Electric and magnetic field patterns evolve across different moments, with a brief radiative phase before the resonance stabilizes [28].

Our focus shifted toward understanding how different geometric configurations could influence and potentially enhance the excitation of anapole states within this material framework [29,30]. We compared the behavior of different free-standing nano-structures: nano-hexagon, nano-octagon, nano-decagon, and nano-ellipsoid of similar dimensions (L = 190 nm, w = 160 nm, and h = 100 nm), to understand how these geometric configurations affect the excitation of anapole states (see Section 4 in Supplement 1). Our results show that while the nano-rectangle and other structures can excite the anapole state, the nano-ellipsoid cannot, despite having similar dimensions.

The impact of geometry on the multipolar contributions to scattering and the phase relationships between different dipole moments is significant, indicating that geometry dictates the appearance of anapole states. For instance, we observe distinct scattering and phase characteristics in the case of various TiO_2 nano-structures (Figs. S5 and S6 in Supplement 1). This finding suggests that the lack of edges in the nano-ellipsoid can be compensated by an increase in size (L = 200 nm, w = 180 nm, and h = 130 nm), emphasizing the significance of geometry in determining anapole states. Figure S4 in Supplement 1 contains a visual representation comparing the nano-rectangle and the nano-ellipsoid, as well as the simulation results of the electric and magnetic fields of the anapole resonance for the different nano-structures.

We have demonstrated so far different types of individual nano-structures supporting anapole excitation in the UV. However, any experimental implementation would require the combination of many single nano-structures for the excitation of the anapoles to be detectable or performed over large areas. In order to demonstrate the feasibility of our single nano-structure designs, we propose a metasurface arrangement based on realistic scales achievable via common nanofabrication techniques.

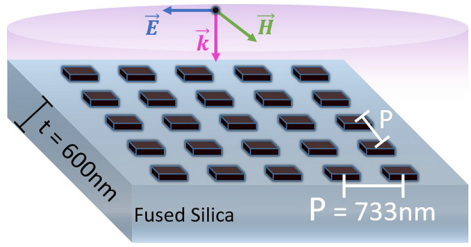


Fig. 4. Metasurface of TiO_2 nano-rectangles deposited atop a fused silica substrate, illuminated by UV light incident perpendicular to the metasurface plane.

Figure 4 illustrates a proposed metasurface formed by amorphous TiO_2 nano-rectangles (length L = 190 nm, width w = 160 nm, and height h = 100 nm deposited atop a fused silica substrate 600 nm thick), illuminated by UV light incident perpendicular to the metasurface plane. The refractive index data for fused silica is based on the work of Malitson (1965) [31]. It is crucial to note that the fused silica substrate is designed to be suspended in order to fulfill fabrication requirements on a possible experimental implementation. The presence of a low refractive index substrate will impact the final response of the metasurface, particularly in the excitation of the anapoles. Therefore, all the following calculations were performed taking into account a thin silica substrate. If the metasurface were embedded in a homogeneous medium, the refractive index contrast would change, potentially affecting the conditions for anapole excitation. The choice of utilizing nano-rectangles in the metasurface design is motivated by our goal to provide realistic and feasible designs. Firstly, results obtained for the nano-rectangle in previous simulations indicate its capability to excite anapoles effectively. Additionally, the nanofabrication techniques that would be required to produce the metasurfaces proposed here (e.g., e-beam lithography) are capable of producing more accurate square and rectangular features than round shapes. Hence, the practicality and ease of fabrication of the nano-rectangle structure, compared to other nano-structures, play a significant role in the decision-making process. All calculations are performed under the same illumination conditions as for the single configuration shown in Fig. 4 with the anapole excitation beam propagation normal to the metasurface and electric field polarization along the longest direction of the rectangular structures.

As can be observed in Fig. 5, we obtain anapole excitation at a wavelength of 370 nm for a periodicity (**P**) between individual meta-atoms of 733 nm. The anapole excitation is shown in Fig. 5(a) (blue curve) as a distinctly narrow dip in the transmittance spectrum characteristic of the anapole state. This significant reduction in scattering indicates that energy should be predominantly trapped near the nano-rectangle arrangement. Calculations for electric and magnetic field intensity at resonant wavelengths corroborate the anapole excitation as shown in Figs. 5(b) and 5(c), where strong field confinement within the nano-rectangle boundaries is observed. Figure S8 in Supplement 1 contains the field intensity maps of the x–z plane cross section of the nano-rectangle.

The anapole formed in the metasurface achieves a Q-factor of 34, approximately. This suggests moderate confinement between electromagnetic fields and radiation losses in our structure. This value may be attributed to the inherent dielectric losses of TiO_2 in the ultraviolet spectrum. We believe that the absorption characteristic of TiO_2 contributes to increased radiation losses,

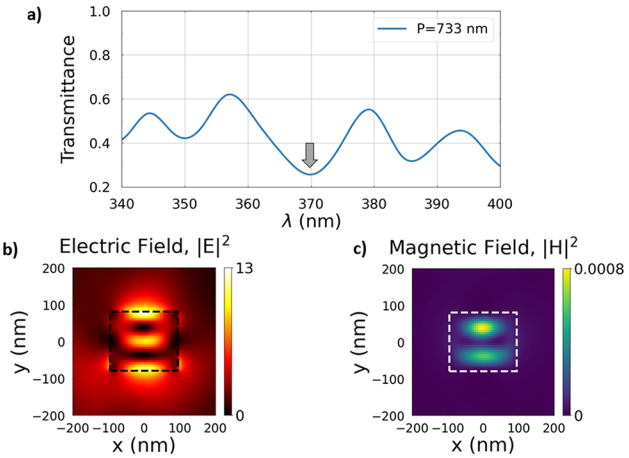


Fig. 5. (a) Transmittance spectrum shows a dip at 370 nm (indicated with an arrow), corresponding to the anapole state for a period of 733 nm (blue curve), indicating strong localization of the electromagnetic field. (b), (c) Field intensity maps showing the electric and magnetic fields, respectively, at the resonant anapole condition (P = 733 nm). The electric and magnetic fields are normalized relative to the source.

thus limiting the Q-factor. Further optimization of the metasurface could potentially enhance the Q-factor by reducing these losses.

It is important to note the role of the periodicity between individual meta-atoms in exciting the anapole state. To ensure that the resonance observed at 370 nm is indeed an anapole state, we conducted simulations for different periods beyond the UV range. Supplement 1 presents results for various periods, showing, for instance, that a period of 700 nm does not effectively excite anapoles. The excitation of the anapole at exactly 733 nm can be explained by taking into account the coupling conditions to in-plane modes of the metasurface.

According to the grating equation [32], a normal incidence beam will diffract into diffraction orders grazing to the grating surface for $\lambda = P/m$ with $m = 1, 2, \ldots$ and where m are the different diffracting orders and λ is the illuminating wavelength. In our case, the anapole excitation occurs at $\lambda = 370$ nm (Fig. 5(a)) which corresponds to $\lambda \approx P/2 \approx 370$ nm, indicating that the second-order diffraction ($m = 2$) plays a significant role in this process. The presence of the second-order diffraction contributes to the field confinement within the nano-rectangles. This diffraction enhances the coupling to the resonance, thereby promoting the strong field confinement that is characteristic of anapoles. A change in period for the same meta-atom dimensions (see Fig. S9 in Supplement 1) prevents efficient anapole excitation, as it opens a channel for de-confinement of the anapole field via diffraction into free space at $\lambda = 370$ nm.

In this study, we demonstrated the excitation of electric anapole states in the UV range using amorphous TiO_2 nano-rectangles. By meticulously tailoring the geometry of these structures, we achieved a non-radiating anapole resonance at approximately 370 nm. Our findings indicate that the anapole resonance is highly sensitive to the geometric parameters of the nano-rectangles and the periodicity of the metasurface. The results emphasize the potential of TiO_2 nano-structures manipulating UV light, with promising applications in fields such as nonlinear optics and photonic devices. The successful excitation of anapole states at short wavelengths opens new avenues for controlling light–matter interactions at the nanoscale, contributing to the advancement of nanophotonics and optical metamaterials.

Funding. Horizon Europe Framework Programme (101057510).

Disclosures. Funded by the European Union. Views and opinions expressed are, however, those of the author(s) only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

Data availability. Data underlying the results presented in this paper are not publicly available but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

REFERENCES

1. C. F. Bohren and D. R. Hoffman, *Absorption and Scattering of Light by Small Particles* (John Wiley and Sons, Ltd., 1998), Chap. 1, pp. 1–11.
2. M. Kerker, H. W. Chew, P. J. Mcnulty, *et al.*, J Histochem Cytochem. **27**, 250 (1979).
3. T. Ergin, N. Stenger, P. Brenner, *et al.*, Science **328**, 337 (2010).
4. I. B. Zel'Dovich, Sov. Phys. JETP **6**, 1184 (1958).
5. J. Wu, F. Zhang, Q. Li, *et al.*, Opt. Express **28**, 570 (2020).
6. K. V. Baryshnikova, D. A. Smirnova, and B. S. Luk'yanchuk, Adv. Opt. Mater. **7**, 1801350 (2019).
7. R. M. Saadabad, L. Huang, A. B. Evlyukhin, *et al.*, Opt. Mater. Express **12**, 1817 (2022).
8. H. B. Sedeh and N. M. Litchinitser, Nanophotonics **12**, 2687 (2023).
9. G. Grinblat, Y. Li, M. P. Nielsen, *et al.*, Nano Lett. **16**, 4635 (2016).
10. G. Grinblat, Y. Li, M. P. Nielsen, *et al.*, ACS Nano **11**, 953 (2017).
11. K. Koshelev, G. Favraud, A. Bogdanov, *et al.*, Nanophotonics **8**, 725 (2019).
12. L. Xu, M. Rahmani, K. Kamali, *et al.*, Light: Sci. Appl. **7**, 44 (2018).
13. M. Semmlinger, M. Zhang, M. L. Tseng, *et al.*, Nano Lett. **19**, 8972 (2019).
14. E. Díaz-Escobar, T. Bauer, E. Pinilla-Cienfuegos, *et al.*, Light: Sci. Appl. **10**, 204 (2021).
15. A. B. Evlyukhin, T. Fischer, C. Reinhardt, *et al.*, Phys. Rev. B **94**, 205434 (2016).
16. J. F. Algorri, D. C. Zografopoulos, A. Ferraro, *et al.*, Nanomaterials **9**, 1 (2019).
17. L. Wei, Z. Xi, N. Bhattacharya, *et al.*, Optica **3**, 799 (2016).
18. K.-H. Kim and W.-S. Rim, ACS Photonics **5**, 4769 (2018).
19. K.-H. Kim, Phys. Chem. Chem. Phys. **22**, 7300 (2020).
20. L. V. R. de Marcos, J. I. Larruquert, J. A. Méndez, *et al.*, Opt. Mater. Express **7**, 989 (2017).
21. C. F. Bohren and D. R. Hoffman, *Absorption and Scattering of Light by Small Particles* (John Wiley and Sons, Ltd., 1998), Chap. 11, pp. 286–324.
22. D. G. Baranov, D. A. Zuev, S. I. Lepeshov, *et al.*, Optica **4**, 814 (2017).
23. A. Jolivet, C. Labbé, C. Frilay, *et al.*, Appl. Surf. Sci. **608**, 155214 (2023).
24. T. Hinamoto and M. Fujii, OSA Contin. **4**, 1640 (2021).
25. A. Miroshnichenko, A. Evlyukhin, Y. F. Yu, *et al.*, Nat. Commun. **6**, 1 (2015).
26. S.-Q. Li and K. B. Crozier, Phys. Rev. B **97**, 1 (2018).
27. Y. Li, G. Zhang, Y. Tang, *et al.*, Nanophotonics **11**, 2245 (2022).
28. A. Canós Valero, E. Gurvitz, F. Benimetskiy, *et al.*, Laser Photonics Rev. **15**, 2100114 (2021).
29. M. Jakob, H. Levanon, and P. V. Kamat, Nano Lett. **3**, 353 (2003).
30. I. Tanabe and Y. Ozaki, J. Mater. Chem. C **4**, 7706 (2016).
31. I. H. Malitson, J. Opt. Soc. Am. **55**, 1205 (1965).
32. J. Quan, Q. Zhou, Y. Cao, *et al.*, Front. Mater. **9**, 1 (2022).